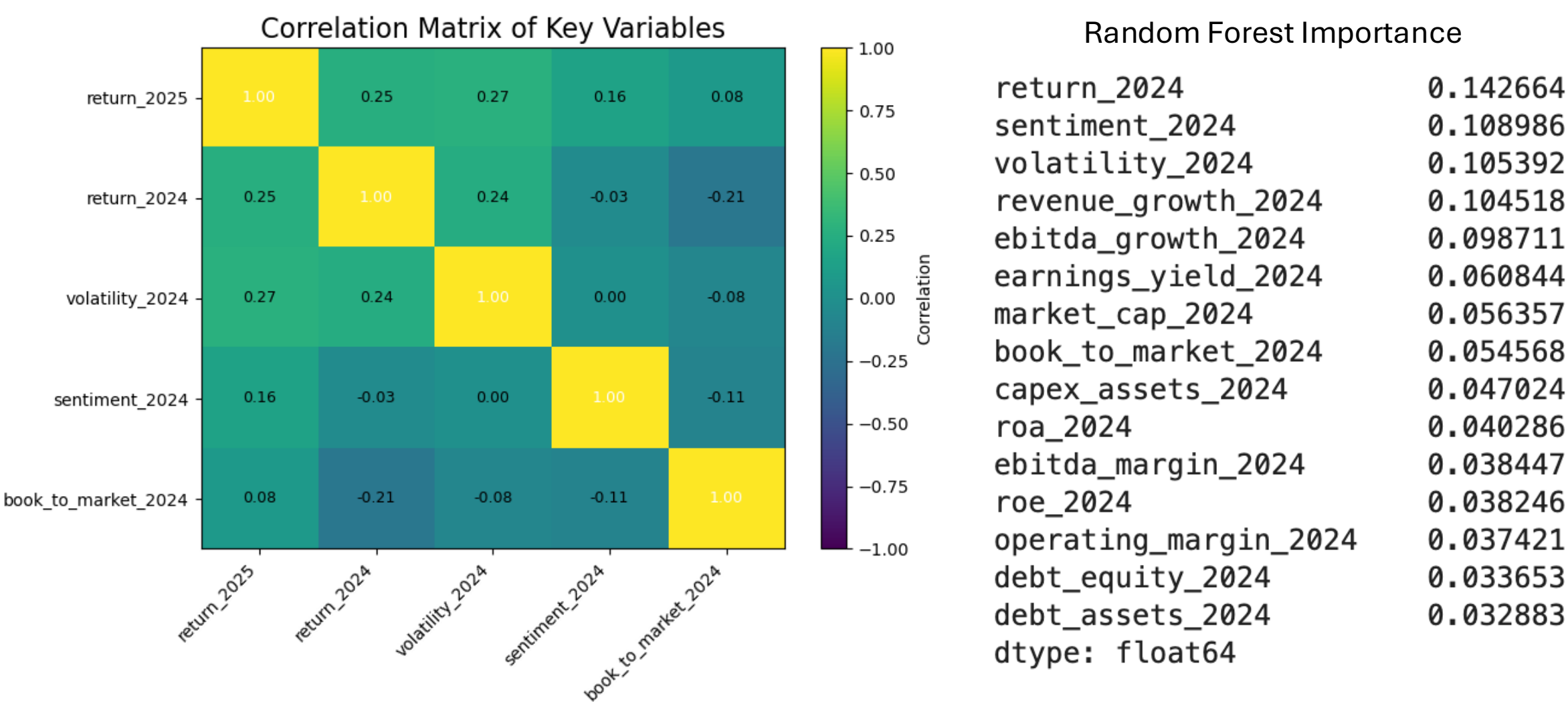


Predicting Equity Returns: Integrating Financial Ratios and Sentiment

Will Ferris and Charlie Baughan | Dickinson College | Data 400 / Professor Bilen

Problem + Data + Method

- Motivation**
- Can we beat the market using data driven insights?
 - What factors can we select to optimize selection?
 - How does sentiment play a role in this?
- Data**
- 2024 S&P 500 constituents
 - Financial indicators: Size, Leverage, Assets, Volatility, Profitability, etc.
 - Source: Bloomberg Terminal
- Data Cleaning & Feature Engineering**
- Normalized data tickers and merged datasets on “Ticker”
 - Dropped duplicate variables and converted numerical columns
 - Removed variables with many missing values
 - Return_2025, 2024 Book to Market, 2024 Earnings Yield
- Methodology**
- Choose the best 2024 predictors of 2025 returns
 - Correlation to returns and Random Forest importance
 - Predictors = 2024 returns, volatility, and sentiment
 - Standardized predictors into a z-score in our manual method and used StandardScaler() in Random Forest to scale and compare



- Scoring system:
 - 0.4(2024 Volatility) + 0.4 (2024 Returns) + 0.2(2024 Sentiment)

- Model**
- Uses the selected predictors to rank stocks
 - Output is ml_score

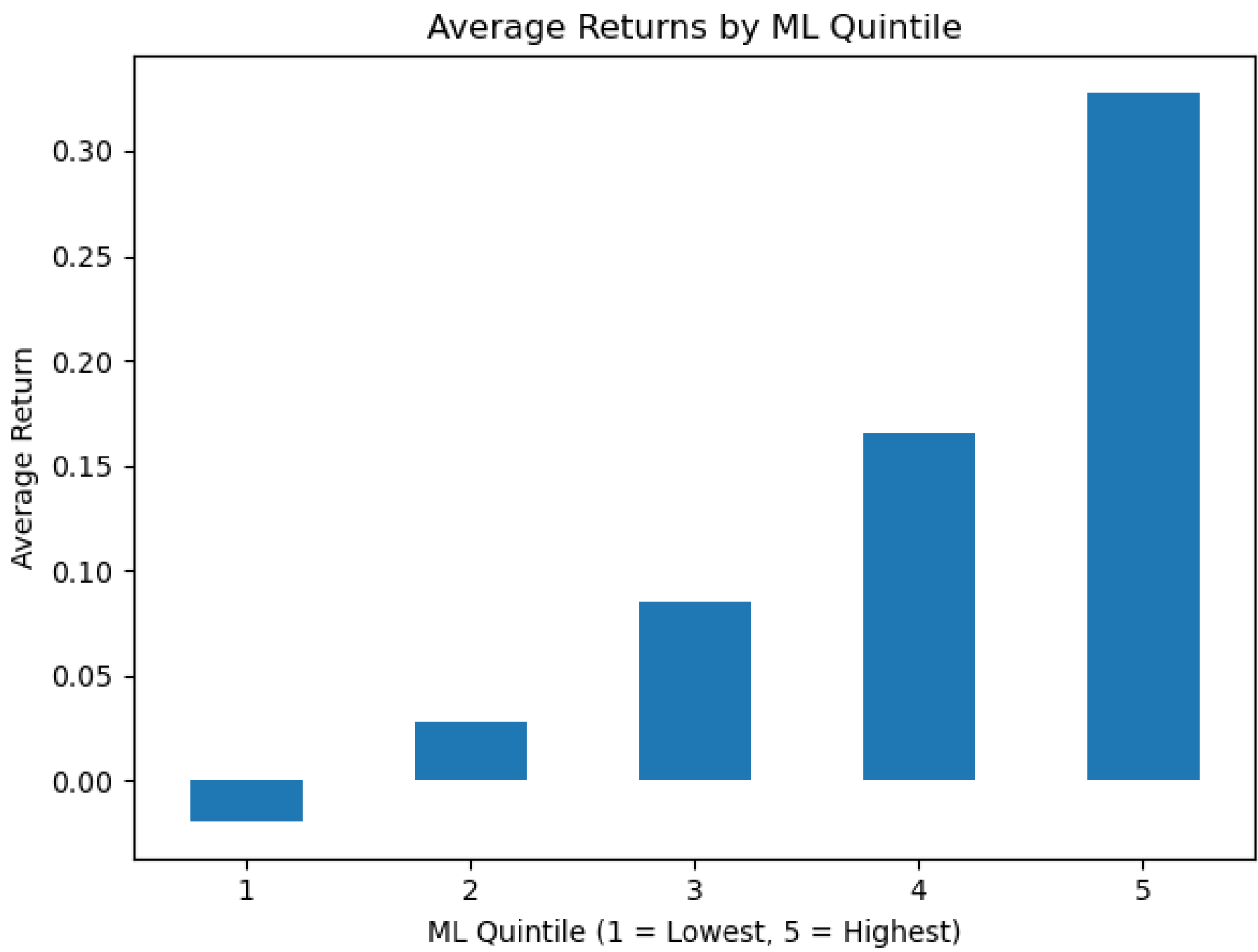
Results

Scoring System Results (top 10)

	ticker	company_name	sector	ml_score
39	APP	APPLOVIN CORP-CLASS A	Information Technology	1.889438
355	PLTR	PALANTIR TECHNOLOGIES INC-A	Information Technology	1.044950
86	CVNA	CARVANA CO	Consumer Discretionary	1.025641
208	GEV	GE VERNOVA INC	Industrials	0.776109
113	CL	COLGATE-PALMOLIVE CO	Consumer Staples	0.646891
394	HOOD	ROBINHOOD MARKETS INC - A	Financials	0.631208
206	IT	GARTNER INC	Information Technology	0.594972
489	WY	WEYERHAEUSER CO	Real Estate	0.580556
101	CIEN	CIENA CORP	Information Technology	0.548929
385	RL	RALPH LAUREN CORP	Consumer Discretionary	0.535392

- Performance Comparison:**
- Stock selection average return: 32.7%
 - Benchmark Index Returns: 10.5%

Score Quintiles by Average Return



- Quintile Machine Learning Spread: 34.72%
- Statistically Significant T – Statistic: 5.67
- Hit Rate: 70% of our selected stocks had positive returns

Interpretation + Strategy

- Key Insights and Applications**
- A systematic, data driven approach can identify predictive signals in equity markets
 - Portfolio sorting by model score produced clear and consistent separation in returns across quintiles
 - Combining financial fundamentals and sentiment data enhances stock selection beyond traditional methods
 - The framework can be applied in practice to:
 - Long only strategies (longing the top-quintile)
 - Long-short strategies(long top-quintile, short bottom-quintile)
- Limitations**
- Analysis based on a single year of data, increasing the risk of overfitting
 - Model weights are derived from limited historical observations and may not generalize
 - Transaction costs and market frictions not incorporated
 - No attempt of diversification made just yet
- Conclusion**
- This study provides evidence that data driven investments can improve stock selection and returns
 - A simple factor-based scoring model can generate meaningful return differentiation across portfolios
 - Integrating sentiment with traditional financial ratios offers incremental predictive power
 - With additional data and more advanced modeling, the framework has strong potential for real world application

- Future Work**
- Extend analysis across multiple years and market cycles
 - Incorporate more machine learning models such as gradient boosting
 - Test sector specific models
 - Evaluate performance after transaction costs and turnover constraints

